

Body Organization & Homeostasis — Recap

Students will describe the levels of structural organization of the human body and explain how homeostasis is maintained through negative feedback.

 Graded online lesson · 1.7: [Organization of Living Things](#)

01 · PROJECT Launch deck — Body Organization & Homeostasis

HOOK

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You are holding steady right now

- ◆ Your temperature, blood sugar, and water are all being held near a **set point** — automatically.
- ◆ Anatomy asks *what* you're made of; physiology asks *how* it works.
- ◆ Today: the 6 levels you're built from, and the loop that keeps you stable.

NEW CONCEPT

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Two words that split the whole course

- ◆ **Anatomy** = structure — what a part IS.
- ◆ **Physiology** = function — what a part DOES.
- ◆ Every unit this year is one of those two questions about a body system.

Today's words

Tissue — a group of similar cells with one job

Organ — two or more tissues working together

Organ system — organs that share a function

Homeostasis — a stable internal environment

Six levels, small to large

- ◆ Chemical → **Cell** → **Tissue** → **Organ** → **Organ system** → **Organism**.
- ◆ Each level is built from the one below it.
- ◆ A tissue = same cells; an organ = different tissues together.

Homeostasis = holding the set point

- ◆ The body keeps conditions (temp, water, glucose) near a **set point**.
- ◆ Drift off it, and a **negative feedback** loop pulls you back.
- ◆ Four parts: **stimulus** → **receptor** → **control center** → **effector**.

Negative doesn't mean bad

COMMON MISTAKE

'Negative feedback' means the response goes in the OPPOSITE direction of the change — it REVERSES it. Sweating when hot is negative feedback, and it's good for you.

KEEP IN MIND

Negative = reverses. Positive = amplifies (rarer, e.g. childbirth).

YOU TRY

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Your turn

YOU TRY

- Order: organ, cell, organism, tissue, organ system, chemical.
- Is shivering negative or positive feedback?

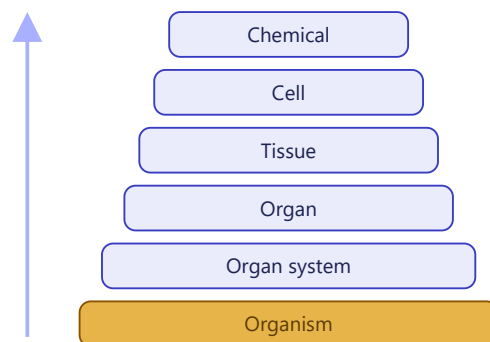
Guided notes — teacher key (students get blanks, printed 2-up)

Body Organization & Homeostasis

TEACHER KEY · Sep 2

Word bank **Anatomy:** structure — what a part IS **Physiology:** function — what a part DOES
Homeostasis: a stable internal environment **Negative feedback:** a response that REVERSES the change

LEVELS OF ORGANIZATION (small → large)



Six levels — each is built from the one below it

Anatomy vs Physiology — anatomy is **structure** (what it is); physiology is **function** (what it does)

Tissue — a group of **similar cells** working together on one job

Organ — two or more **tissues** working together (e.g. the heart)

Homeostasis — keeping the body's internal conditions **stable** despite outside change

CONCEPT 1 — THE SIX LEVELS OF ORGANIZATION

How does the body build from atoms up to a whole person?

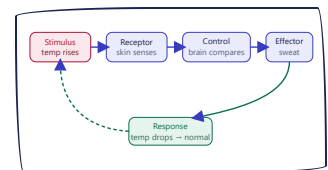
1. Atoms join into molecules — the **chemical** level.
2. Molecules build a **cell**, the smallest unit of life.
3. Many similar cells form a **tissue**; several tissues form an **organ**.
4. Organs that share a job form an **organ system**; all systems together make the **organism**.

chemical → cell → tissue → organ → organ system → organism

CONCEPT 2 — HOMEOSTASIS BY NEGATIVE FEEDBACK

You get too hot. How does your body pull you back to normal?

1. A change pushes a variable off its set point — the **stimulus**.
2. A **receptor** detects the change and sends the signal in.
3. The **control center** (often the brain) compares it to the set point.
4. An **effector** makes a response that **reverses** the change — bringing you back to normal.



stimulus → receptor → control center → effector → response (reverses the change)

CHECK YOUR UNDERSTANDING

1.

Put in order, smallest to largest: organ, cell, organism, tissue, organ system, chemical.

chemical, cell, tissue, organ, organ system, organism

2.

Shivering when you get cold — negative or positive feedback? Explain in one sentence.

Negative feedback — shivering makes heat to REVERSE the drop in temperature, back toward the set point.

3.

Why is the heart an organ and not a tissue?

It is made of two or more different tissues working together, which defines an organ.

SUMMARY

Anatomy is **structure**, physiology is **function**. The body builds **chemical to organism**, and **negative feedback** keeps it stable by reversing changes.

Worksheet — Body Organization & Homeostasis — Recap

Use your notes. These are the exact terms and ideas your quiz will ask.

TYPE 1 Label & order

Directions: Label the pyramid with the six levels in order (smallest at top), using the word bank.

1. Level 1 (smallest):
2. Level 3:
3. Level 6 (largest):

TYPE 2 Match term to function

Directions: Write the term from the word bank that each description defines.

4. A group of similar cells doing one job:
5. The study of what a body part DOES:
6. Keeping the internal environment stable:
7. A control loop whose response reverses the change:

TYPE 3 Explain & apply

Directions: Answer in one or two sentences.

8. Why is the stomach an organ and not a tissue?
9. A runner's body temperature climbs during a race, then their sweat glands kick in and it drops back. Name each part of this feedback loop (stimulus, receptor, control center, effector) and say why it is NEGATIVE feedback.

Answer key

LABEL & ORDER

1. Chemical
2. Tissue
3. Organism

MATCH TERM TO FUNCTION

- 4. Tissue
- 5. Physiology
- 6. Homeostasis
- 7. Negative feedback

EXPLAIN & APPLY

- 8. The stomach is made of several different tissues (muscle, lining/epithelium, nerve) working together, and 'organ' means two or more tissues combined.
- 9. Stimulus = body temp rises; receptor = skin/hypothalamus senses it; control center = hypothalamus (brain) compares to the set point; effector = sweat glands produce sweat. It is **NEGATIVE** feedback because the response (cooling) reverses the original change (heating).

06 · CLOSE **Exit ticket**

- 1. Put the 6 levels of organization in order (smallest to largest).
- 2. Name the 4 parts of a negative feedback loop, in order.